

Getting Started with Pygame



Part—2

In the first part of this series, we learnt about uploading a background image, and how to put text with different fonts on the screen. We also went through event-handling. In this part, we are going to learn about some important aspects such as colours, getting input from the user, rotation using the mouse and keyboard, and my favourite: sprites.

The first thing that came to mind when thinking of colours was RGB (red, green, blue). These are the primary colours, and you can create every colour out of these three by just varying the intensity of each one. In Part 1 of this series, to pass a colour as an argument, I simply passed a tuple of three elements, like (255,0,0). The first element in the tuple corresponds to the intensity of the red colour, the others to green and blue. The intensity ranges from 0 to 255. Brightness increases as you move from 0 to 255. (0,0,0) is black, and (255,255,255) yields white. Given below is a little script that accepts intensities for the three colours from the user, and displays the resultant colour. (Find script and image files on the LFY website at http://linuxforu.com/article_source_code/july11/pygame.zip.)

```
#color.py
import pygame
from pygame.locals import *
from sys import exit

r=min(255, input("Enter the red colour intensity :"))
```

```
g=min(255, input("enter the green colour intensity :"))
b=min(255, input("enter the blue colour intensity :"))
screen=pygame.display.set_mode((640,480),0,24)
pygame.display.set_caption("colour testing ")
```

```
while True:
    for i in pygame.event.get():
        if i.type==QUIT:
            exit()
        screen.fill((r, g, b))
        pygame.display.update()
```

We have used the *min* function to limit user-entered values to a maximum of 255. The *screen.fill()* function fills the screen with the colour denoted by the (r, g, b) tuple passed as the argument. For example, values of (123, 212 and 255) will give a colour as shown in Figure 1.

Blending

Suppose you have a green character in your game and, suddenly, a yellow fireball hits him. The character's colour